Introduction to Nutrition and Human Physiology

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Introduction to Nutrition and Human Physiology (A33081)

Short Title:	Intro Nutr & Human Phys		
Faculty	Science and Computing		
Department	Science		
Cluster	Food Science		
Author	EDUGGAN		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module introduces the study of nutrition and the basics of human physiology. The student will gain an understanding of the fundamentals of the functioning of the human body from homeostasis to the functioning of cells and the various body systems. The student will also develop their knowledge in the role of nutrients in human health and evaluate diets using current methods.

Programmes

`SCIE-0087`	BSc (Hons) in Food Science and Innovation (`WD_SFSIN_B`)	2 / 3 / M
	BSc in Food Science with Business (`WD_SFDSC_D`)	2 / 3 / M

Indicative Content

- Basics of nutrition
- Fundamentals of human physiology including homeostasis and cellular physiology.
- Structure and functions of the various body systems with nutritional significance including the cardiovascular, nervous, digestive, hepatic, renal, respiratory, muscular and immune systems.
- Nutrients in food and their role in health: proteins, fats, carbohydrates, vitamins, minerals.
- Food habits and their impact on nutritional intake.
- Dietary analysis and assessment of dietary intake.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Define nutrition and discuss the basics of a healthy diet.
2. Explain homeostasis and cellular physiology as related to nutrition.
3. Describe the structure and functions of body systems and their role in maintaining homeostasis.
4. Explain the role of nutrients in health.
5. Discuss the influences of food habits on nutritional intake.
6. Evaluate diets using a dietary analysis programme.

Learning and Teaching Methods

- Lectures
- Case studies
- Group discussions and presentations
- Dietary analysis computer programmes

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Continuous Assessment	30%	
Assignment	30%	1,6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks. Failure to meet the objectives of projects. Unable to carry out or submit independent study and research.

Note: In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Notes:

- 40%-49% Demonstrate a limited knowledge of the subject matter. Shows some technical ability/skill in carrying out and reporting on practical/professional tasks.
- 50%-59% Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.
- 60%-69% Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argues in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks. Projects and reporting coherent and soundly structured illustrating wide and in-depth reading on the subject coupled with the proper use of references.
- 70%-100% Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	12	
Independent Learning	99	

Essential Material

- "Moodle." moodle.wit.ie
- Barasi, M.E. _Human Nutrition: A healthy perspective_. 2nd ed. London: Oxford University Press, 2003.
- Sherwood, L. _Human physiology: from cells to systems_. 5th ed. UK: Thomson Brooks/Cole, 2004.

Supplementary Material

- "Journal of Human Physiology." https://ojs.bilpublishing.com/index.php/jhp
- "Journal of Nutrition." https://academic.oup.com/jn

Requested Resources

- Lecture Room: Loose Seated
- COMPUTER LAB: Networks Lab